

TRN004 – Power BI & Advanced Analytics

1. Course Overview

- **Duration:** 2 Days | 4 Hours per Day | Total 8 Hours
- **Delivery Mode:** Online Instructor-Led Training
- **Category:** Analytics | Data Visualization | Business Intelligence
- **Platform Focus:** Microsoft Power BI
- **Target Audience:** Business Analysts, Data Analysts, Reporting Teams, MIS Executives, Decision Makers

2. Course Objective

This training program enables participants to build interactive dashboards, transform raw datasets into business insights, create advanced visualizations, and use analytical techniques within Power BI for enterprise reporting and data-driven decision-making.

3. Learning Outcomes

- Understand the end-to-end Power BI workflow and analytics architecture.
- Connect, clean, and transform data using Power Query.
- Create data models and relationships for business reporting.
- Build interactive dashboards and visual reports.
- Write DAX calculations and advanced business measures.
- Generate insights using analytics and storytelling techniques.
- Publish and share enterprise reports securely.

4. Curriculum Structure

DAY 1 – POWER BI FOUNDATIONS & DATA VISUALIZATION

Session	Session 1: Introduction to Power BI
Duration	45 Minutes
Topics Covered	Power BI Ecosystem, Business Intelligence Concepts, Power BI Desktop, Data Workflow
Lab / Workshop	Power BI Environment Walkthrough
Expected Outcome	Understand the Power BI architecture and reporting process.

Session	Session 2: Data Import & Transformation
Duration	60 Minutes
Topics Covered	Connecting Data Sources, Power Query Editor, Data Cleaning, Data Transformation
Lab / Workshop	Transform Raw Sales Data
Expected Outcome	Prepare clean and structured business datasets.

Session	Session 3: Data Modeling Fundamentals
Duration	60 Minutes
Topics Covered	Relationships, Star Schema, Data Types, Calculated Columns, Hierarchies
Lab / Workshop	Build a Data Model
Expected Outcome	Create optimized data models for reporting.

Session	Session 4: Dashboard & Visualization Design
Duration	75 Minutes
Topics Covered	Charts, KPIs, Tables, Slicers, Drilldowns, Visual Design Best Practices
Lab / Workshop	Interactive Dashboard Workshop
Expected Outcome	Develop business-friendly interactive dashboards.

Session	Session 5: Storytelling with Data
Duration	20 Minutes
Topics Covered	Insight Presentation, Visual Storytelling, Business Reporting Techniques
Lab / Workshop	Dashboard Review Exercise
Expected Outcome	Improve executive reporting and communication.

DAY 2 – ADVANCED ANALYTICS & REPORTING

Session	Session 6: DAX Fundamentals
Duration	60 Minutes
Topics Covered	Measures, Aggregations, Time Intelligence, Calculated Measures
Lab / Workshop	Create Business KPIs using DAX
Expected Outcome	Build dynamic analytical calculations.

Session	Session 7: Advanced Analytics in Power BI
Duration	60 Minutes
Topics Covered	Forecasting, Trend Analysis, Conditional Formatting, AI Visuals
Lab / Workshop	Sales Trend Analytics Exercise
Expected Outcome	Generate predictive and analytical business insights.

Session	Session 8: Report Optimization & Security
Duration	45 Minutes
Topics Covered	Performance Optimization, Row-Level Security, Report Governance
Lab / Workshop	Secure Dashboard Implementation
Expected Outcome	Implement secure and enterprise-ready reports.

Session	Session 9: Publishing & Collaboration
Duration	45 Minutes
Topics Covered	Power BI Service, Workspaces, Sharing Reports, Scheduled Refresh
Lab / Workshop	Publish Reports to Power BI Service
Expected Outcome	Deploy and collaborate using cloud reporting.

Session	Session 10: Final Analytics Capstone
Duration	50 Minutes
Topics Covered	Business Dashboard Design & Executive Insights Presentation
Lab / Workshop	Power BI Analytics Capstone
Expected Outcome	Present complete analytical dashboards with recommendations.

5. Hands-On Labs & Activities

- Import and transform datasets using Power Query
- Create enterprise-grade data models
- Build KPI dashboards and executive reports
- Develop DAX calculated measures
- Perform sales and business trend analysis
- Implement report security and governance
- Publish reports using Power BI Service
- Present business insights through dashboards

6. Assessment Framework

Assessment Component	Weightage
Hands-On Labs	30%
Knowledge Checks & Exercises	20%
Dashboard Workshops	20%
Final Analytics Capstone	30%

7. Final Capstone Project

Participants design a complete business analytics dashboard using Power BI and present analytical insights for executive decision-making.

- Connect and transform raw business data.
- Build business KPIs and advanced measures.
- Design interactive dashboards and analytics views.
- Generate insights and business recommendations.
- Present findings in an executive reporting format.

8. Expected Deliverables

- Power BI Dashboard Project
- Cleaned & Transformed Dataset
- Business KPI Report
- DAX Measures Workbook
- Analytics Insight Presentation

9. Expected Business Outcomes

- Improve decision-making through real-time analytics.
- Enhance business reporting and dashboard automation.
- Increase visibility into operational and sales metrics.
- Enable data-driven culture within teams.
- Improve reporting accuracy and efficiency.
- Strengthen enterprise analytics capabilities.

10. Training Methodology

Theory & Concepts: 35% | **Hands-On Labs & Dashboard Building:** 55% | **Discussions & Business Case Studies:** 10%